Selecting, Not Smoothing: Physically Grounded Object Trajectories from Egocentric Video for Dexterous Robot Imitation

V2D Challenge 2026 — Track 3 Technical Report

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Abstract: We present our Track 3 system, which turns an egocentric stereo recording of a person manipulating kitchen objects into a dexterous robot-hand policy in simulation. Using only our own reconstructions (no provided scans or motion capture), we estimate metric object meshes, hand motion and 6-DoF object trajectories, ground them physically, and train one policy per episode with soft actor-critic. Our main lesson is that object trajectories must be *selected* globally rather than smoothed locally: we keep every FoundationPose hypothesis per frame and choose a whole-sequence path with SAM 3 masks and TrackEverything 3D point tracks, which removes the 180° flips and self-spins that make human motion impossible to imitate. Simulator-side grounding (settled resting poses, collision geometry, hands pushed out of objects) matters as much as pose accuracy. Our best submission reaches AUC 0.584, SP-SR 0.650 and MPPE 6.06 cm.

1. Introduction

Track 3 of the Video to Data challenge [10] asks for an end-to-end transfer: from an egocentric video (colour camera plus a mono stereo pair) a robot policy must re-enact the human manipulation in the Flash-CHORD simulator [9] with floating Sharpa hands, and the objects it moves are scored against hidden ground truth; reconstructed meshes are scored separately (CD-O). Chaining off-the-shelf components fails at the hand–object interface: under hand occlusion and for near-symmetric objects (a spoon, a lid, flat boards) per-frame poses flip by up to 180°, held objects spin relative to the hand, and resting objects start tilted into the table. A robot cannot imitate such references. Recent work tackles the same interface by multi-hypothesis selection [3], contact-aware factor graphs [6], learned hand/object trust [16], agentic generation [15] and diffusion denoising [7]. Our contributions:

- SelectTrack: whole-sequence selection among all FoundationPose [21] hypotheses, scored by hand-aware SAM 3 silhouettes [2] and constrained by TrackEverything [5] 3D point tracks, followed by physically motivated filters;
- simulator grounding: simulator-settled starts, hand depenetration, container-aware collision and data-driven mesh corrections, which turned several never-solved episodes into successes;

2. Approach

Perception. We rectify the published colour-to-stereo extrinsic (3.4 vs. 18.9 px reprojection error), run FoundationStereo [22] for metric depth and DROID-SLAM [17] for ego-motion, and fit the round table plane for gravity. A VLM, Qwen3-VL [1], describes the task from 8 frames and names each object with a short noun phrase (the object identifier is a fallback); these phrases prompt SAM 3 [2], which segments objects and hands (masks match the hand-written prompts within 0.02 IoU on 35/36 objects). Object meshes come from SAM 3D Objects [13] on automatically picked unoccluded views, metrically scaled by stereo depth; the consensus of several samples is kept. Hands are Dyn-HaMR [23] MANO [12] trajectories in metric scale; the few frames it misses (3157392%) fall back to WiLoR [11] keypoints triangulated across the stereo pair.

Object trajectories (SelectTrack). For every frame t we keep all $K \approx 252$ refined FoundationPose hypotheses X_t^k with scores s_t^k, instead of the argmax. TrackEverything [5] supplies surface points $p_n(t)$ tracked in the SLAM world (seeded on object masks minus hand pixels). Following the multi-hypothesis idea of Hand2Robot [3], we decode one hypothesis per frame by Viterbi,

$$\min_{k_{1:T}} \sum_t \Big[1 - \hat{s}_t^{k_t}\,\mathrm{IoU}_t^{k_t} + \lambda_p\, d_{\mathrm{surf}}(X_t^{k_t}) + \lambda_m \textstyle\sum_n \big\| X_t^{k_t} (X_{t-1}^{k_{t-1}})^{-1} p_n(t-1) - p_n(t) \big\| \Big], \tag{1}$$

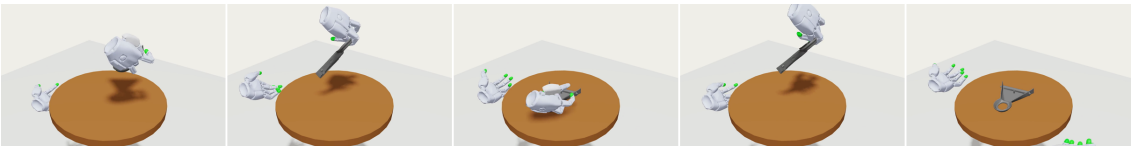


Figure 1: Learned policy re-enacting the dust-pan episode (eval ep. 38): grasp at the ring handle, carry, set down. This episode was never solved before simulator-settled starts and stickier contact (own-reference MP-SR $0 \rightarrow 0.98$).

where IoU ignores hand pixels (a covering hand is not evidence), d_{surf} is the distance of visible tracked points to the hypothesis surface, and the pairwise term demands that consecutive hypotheses move the tracked points as they actually moved; a flip or spin becomes expensive. Tracks are then anchored in the object frame so that points seen far apart in time land on the same mesh location. Physical filters follow: a speed-limit Viterbi that keeps, flips or drops frames ($\leq 20°$, 6 cm per frame), a rest hold for untouched objects, a spin guard letting held objects rotate only with the holding hand plus slip (inspired by [6, 16]), and exact twist unwrapping for rotationally symmetric cups. A VICON-style soft pose graph [6] improved mocap accuracy slightly but cannot escape a 180° flip; we also considered [8, 18, 24].

Simulator grounding. Reconstructed fingers sat 2–4 cm inside cups, so reset shoved hand and cup apart; we translate each hand (time-smoothed) out of every object. Resting objects carried a $\sim 12°$ tilt about the camera's horizontal axis—the least observable direction—and tipped at reset; we drop each scene into the simulator for 2 s without the robot and adopt the settled poses as initial rests (accepted if $\leq 20°/3$ cm). Collision is CoACD [20] on 4 mm walls (openings stay open), a dish-rack slot is carved where the sweeper really stood, and cups get a solid base. The pitcher mesh was lengthened at its base from a measurement (it rested 36 mm above the table in all three pitcher episodes).

Policy learning. One FlashSAC [4, 9] policy per episode (2048 worlds, 250 M steps, real-time playback) with the reward $r = 40\, r_{\text{obj}} + 5\, r_{\text{cloc}} + 10\, r_{\text{wrench}} + 5\, r_{\text{prog}} + 200\, \mathbb{1}_{\text{done}} - \ldots$, i.e. object keypoint tracking, a contact-aware term rewarding robot fingers at the human contact locations and contacts able to support the object, progress while every object is within 3 cm/0.3 rad, and a completion bonus, with termination limits annealed from 3× to the official values. We train two seeds and a stickier contact (friction 1.5) and submit, per episode, the run with the best own-reference MP-SR, then SP-SR, then AUC.

Submission	AUC↑	SP↑	MP↑	RPE↓	MPPE↓	CD-O↓
SelectTrack + pose filter	0.504	0.50	0.15	9.23	8.60	0.407
+ physical filters, grounding	0.532	0.55	**0.20**	8.10	7.82	0.407
+ 2nd seed	0.553	0.60	0.15	7.54	7.18	0.401
+ settled starts, friction	**0.584**	**0.65**	0.15	**7.35**	**6.06**	**0.401**
leader	0.765	0.80	0.50	3.85	3.00	–

Table 1: Official test scores (20 evaluation episodes, own meshes). RPE, MPPE, CD-O in cm.

3. Experiments and Results

Table 1 shows the official scores of successive submissions: every component improved AUC, SP-SR, RPE and MPPE (about 12th of 37 by summed rank). SAC clearly outperformed PPO [14] on identical data and reward (median reference progress 0.62 after 250 M steps vs. 0.27 after 492 M; App. A). Simulator settling cut the mean rotation of resting objects during the first 3 s from 12.3° to 5.0° and made the dust-pan, sweeper and board episodes solvable (Fig. 1). Two findings were negative: rigidly locking held objects to the reconstructed palm lowered mocap accuracy (0.87→0.55–0.72 of carried frames within 15°/3 cm), because the palm frame drifts and objects genuinely slip; and hand-only refinement [7] cannot repair object poses. Most tellingly, 16 of 20 episodes pass MP-SR against our own reconstruction but only 3 against the hidden ground truth.

4. Conclusion

Global hypothesis selection with point-track consistency, plus simulator-side grounding, makes egocentric human motion imitable by a dexterous robot hand; the remaining gap is reconstruction accuracy, especially the camera-depth tilt of resting objects and centimetre offsets of held ones. Feeding simulator feedback back into the reconstruction (as in DeepSimHO [19]) and refining meshes jointly with poses are the most promising next steps. Code: https://github.com/KiaRational/v2d-track3.

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A. Additional Results

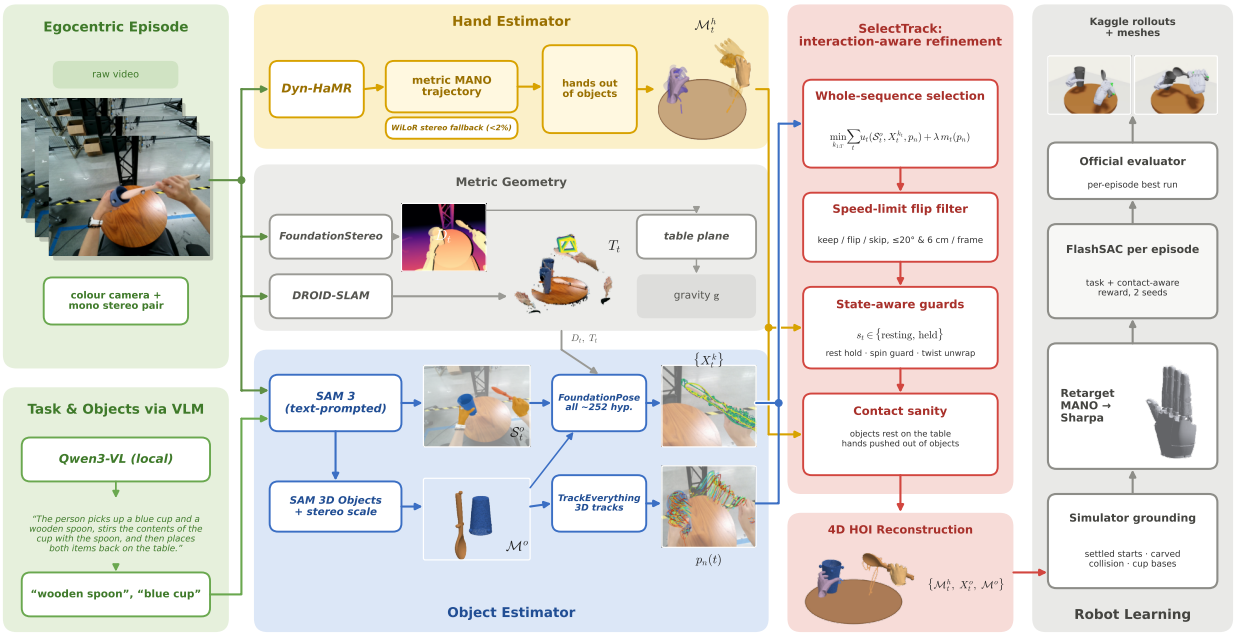


Figure 2: System architecture. A VLM names the objects for SAM 3; hand and object estimators produce MANO hands, meshes, all FoundationPose hypotheses and 3D point tracks; SELECTTRACK selects and grounds object trajectories; the 4D reconstruction is retargeted to Sharpa hands and a per-episode SAC policy is trained in simulation.

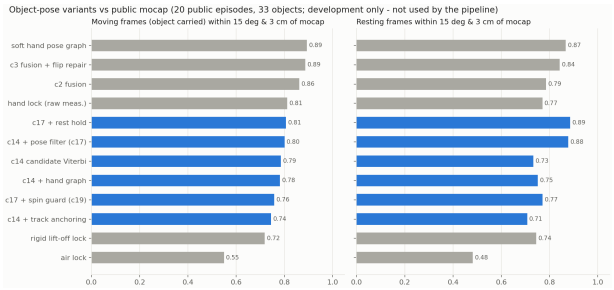


Figure 4: Object-pose variants vs. public mocap (development only). SELECTTRACK trades a little carried-frame accuracy for the removal of flips and spins, which is what the robot needs.

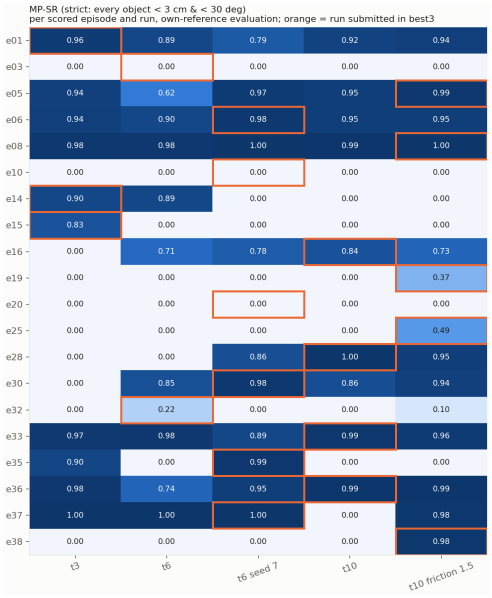


Figure 5: Own-reference MP-SR per scored episode and training run; outlined: run submitted.

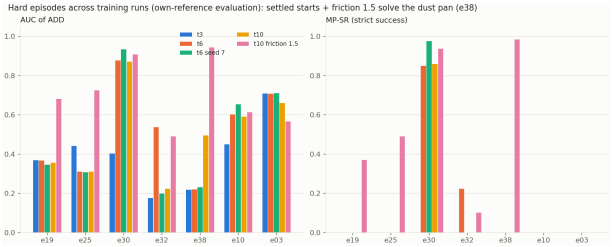


Figure 6: Hard episodes across runs: settled starts and friction 1.5 solve the dust pan (e38) and boards (e25).